

Revenue Debt Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Revenue Debt Discipline (RDD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on RDD achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.62 (2.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Asset prices reflect the compensation investors require for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to states where marginal utility is high. While consumption-based asset pricing theory provides a coherent framework for understanding return predictability, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify Revenue Debt Discipline (RDD) as a novel predictor of stock returns that captures firms' exposure to aggregate consumption dynamics. Firms exhibiting greater revenue debt discipline—measured as the negative ratio of change in long-term debt to sales—demonstrate systematically different return patterns that align with consumption-based pricing mechanisms. This characteristic reflects firms' financial flexibility and their ability to smooth cash flows during periods of consumption volatility, suggesting a direct link between corporate financing decisions and investors' marginal utility of consumption.

Revenue Debt Discipline captures firms' exposure to consumption risk through their differential sensitivity to intertemporal marginal rate of substitution. Firms with high RDD maintain conservative debt policies relative to revenue generation, providing a buffer against consumption shocks that affect investors' marginal utility [Campbell and Cochrane \(1999\)](#). These firms exhibit lower sensitivity to long-run consumption risks, as their financial flexibility allows them to maintain stable operations during periods when consumption growth is uncertain [Bansal and Yaron \(2004\)](#). The consumption-based interpretation suggests that high-RDD firms serve as hedges against bad consumption states, when marginal utility is high, thereby commanding lower risk premia. During economic downturns characterized by declining consumption growth, firms with disciplined debt management relative to revenues can better weather adverse shocks without amplifying investors' consumption risk [Lettau and Ludvigson \(2001\)](#). This state-dependent behavior manifests through the

covariance between firm cash flows and the stochastic discount factor, where high-RDD firms provide relatively higher payoffs precisely when consumption is low and marginal utility is high. The disaster risk framework further reinforces this mechanism, as firms with greater revenue debt discipline maintain operational flexibility that becomes particularly valuable during rare consumption disasters [Barro \(2006\)](#). Through this lens, RDD proxies for firms’ systematic exposure to consumption-based pricing kernels, with the expected return differential reflecting compensation for bearing aggregate consumption risk.

We find strong evidence that Revenue Debt Discipline predicts cross-sectional stock returns. The value-weighted long/short RDD strategy achieves an annualized gross Sharpe ratio of 0.46 and generates a monthly average return of 23 basis points with a t-statistic of 3.27. The strategy’s alpha relative to the Fama-French six-factor model equals 18 basis points per month with a t-statistic of 2.62, demonstrating robustness to known risk factors. The RDD strategy loads significantly on the CMA factor with a coefficient of 0.35 and t-statistic of 7.53, consistent with the consumption-based interpretation that disciplined firms require lower returns. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the RDD strategy delivers an average return of 29 basis points per month with a t-statistic of 3.18. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically significant at 0.34, placing the strategy in the top 7% of documented anomalies. Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the RDD strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.24. The strategy’s performance persists across alternative portfolio construction methodologies, with nineteen out of twenty-five specifications yielding t-statistics exceeding three for gross alphas.

Our study contributes to the consumption-based asset pricing literature by iden-

tifying Revenue Debt Discipline as a characteristic that captures cross-sectional variation in consumption risk exposure. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) establish the theoretical and empirical foundations for consumption-based explanations of the cross-section, we provide direct evidence linking corporate financial policies to consumption risk premia. Our findings extend [Zhang \(2005\)](#) by demonstrating that financing decisions, particularly the interaction between debt and revenue dynamics, contain information about firms’ systematic risk exposures beyond traditional investment-based measures. Unlike [Cooper et al. \(2008\)](#), who focus on asset growth as a predictor, we show that the discipline in debt management relative to revenue generation provides a more refined measure of firms’ exposure to aggregate consumption fluctuations. Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish that RDD’s predictive power survives stringent robustness tests including transaction costs and controls for related anomalies. The RDD effect remains significant after controlling for established predictors such as net debt financing, asset growth, and employment growth, suggesting it captures a distinct dimension of consumption risk. Our results have broader implications for understanding how corporate policies affect required returns through their impact on firms’ covariance with marginal utility, providing a consumption-based rationale for the observed return predictability that complements existing behavioral and limits-to-arbitrage explanations.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data necessary to construct the Revenue Debt Discipline signal. The sample spans multiple decades,

allowing us to examine the signal’s properties across various market conditions and business cycles. Revenue Debt Discipline signal utilizes two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities with maturities exceeding one year. This item reflects the cash inflows from debt financing activities as reported in the statement of cash flows. Sales (SALE) represents the total revenue generated from a firm’s primary business operations during the fiscal year, providing a measure of the firm’s operating scale and economic activity. We construct the Revenue Debt Discipline signal by calculating the year-over-year change in long-term debt issuance scaled by lagged sales: $(DLTIS(t) - DLTIS(t-1)) / SALE(t-1)$. This measure captures the change in a firm’s debt financing activities relative to its revenue base, providing insight into how aggressively firms are expanding their debt financing relative to their operating scale. A higher value indicates an acceleration in debt issuance relative to the firm’s revenue-generating capacity, while a lower or negative value suggests more disciplined debt management or deleveraging. We compute the signal using annual observations at each fiscal year-end and require non-missing values for long-term debt issuance in consecutive years and lagged sales. To ensure meaningful ratios, we exclude observations where lagged sales are zero or negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the RDD signal. Panel A plots the time-series of the mean, median, and interquartile range for RDD. On average, the cross-sectional mean (median) RDD is -0.47 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input RDD data. The signal’s interquartile range spans -0.12 to 0.15. Panel B of Figure 1 plots the time-

series of the coverage of the RDD signal for the CRSP universe. On average, the RDD signal is available for 6.12% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does RDD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on RDD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high RDD portfolio and sells the low RDD portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short RDD strategy earns an average return of 0.23% per month with a t-statistic of 3.27. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.18% to 0.29% per month and have t-statistics exceeding 2.62 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.35, with a t-statistic of 7.53 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 526 stocks and an average market capitalization of at least \$1,687 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies

are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 16 bps/month with a t-statistics of 3.39. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -10-24bps/month. The lowest return, (-10 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.67. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the RDD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in

nineteen cases.

Table 3 provides direct tests for the role size plays in the RDD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and RDD, as well as average returns and alphas for long/short trading RDD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the RDD strategy achieves an average return of 29 bps/month with a t-statistic of 3.18. Among these large cap stocks, the alphas for the RDD strategy relative to the five most common factor models range from 18 to 34 bps/month with t-statistics between 1.99 and 3.69.

5 How does RDD perform relative to the zoo?

Figure 2 puts the performance of RDD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the RDD strategy falls in the distribution. The RDD strategy’s gross (net) Sharpe ratio of 0.46 (0.34) is greater than 86% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the RDD strategy (red line).² Ignoring trading costs, a \$1 invested in the RDD strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.57 which ranks the RDD strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the RDD strategy would have yielded \$1.54 which ranks the RDD strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the RDD relative to those. Panel A shows that the RDD strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The RDD strategy has a positive net generalized alpha for five out of the five factor models. In these cases RDD ranks between the 69 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does RDD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of RDD with 210 filtered anomaly

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price RDD or at least to weaken the power RDD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on RDD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the RDD signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.19, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on RDD is -0.66.

Similarly, Table 5 reports results from spanning tests that regress returns to the RDD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the RDD strategy earns alphas that range from 15-19bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.22, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the RDD trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does RDD add relative to the whole zoo?

Finally, we can ask how much adding RDD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the RDD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes RDD grows to \$1138.20.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which RDD is available.

8 Conclusion

This study provides robust evidence that Revenue Debt Discipline (RDD) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that RDD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on RDD delivers an annualized gross Sharpe ratio of 0.46, which remains economically significant at 0.34 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 18 basis points gross (13 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds, underscores its statistical reliability. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related strategies from the factor zoo. The persistence of a 15 basis points monthly alpha with a t-statistic of 2.24 in this stringent specification suggests that RDD captures unique information about firm fundamentals not fully reflected in existing pricing factors.

From a practical perspective, these findings have important implications for both asset managers and academic researchers. The net Sharpe ratio of 0.34 indicates that RDD-based strategies remain profitable even after realistic implementation costs, making them potentially attractive for institutional investors. The signal’s distinctiveness from related factors such as net debt financing, inventory growth, and employment growth suggests it may serve as a valuable addition to multi-factor investment frameworks.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, we do not explicitly model market impact or capacity

constraints that might affect the scalability of RDD-based strategies. Second, the study period and market conditions examined may not fully capture the signal's performance across different economic regimes or market stress scenarios. Third, the economic mechanism underlying RDD's predictive power requires further theoretical development to enhance our understanding of why this signal generates abnormal returns.

Future research should explore several promising avenues. Investigation into the time-varying nature of RDD's predictive power could reveal whether the signal's effectiveness depends on macroeconomic conditions or market sentiment. International evidence would help establish whether RDD's pricing implications extend beyond U.S. markets, potentially uncovering market-specific institutional factors that influence its effectiveness. Additionally, examining the signal's performance in different firm size segments and industry classifications could provide insights into where RDD strategies might be most profitably implemented. Finally, developing a more comprehensive theoretical framework linking revenue debt discipline to firm value creation would strengthen the economic foundations of this empirical finding and potentially suggest refinements to the signal's construction.

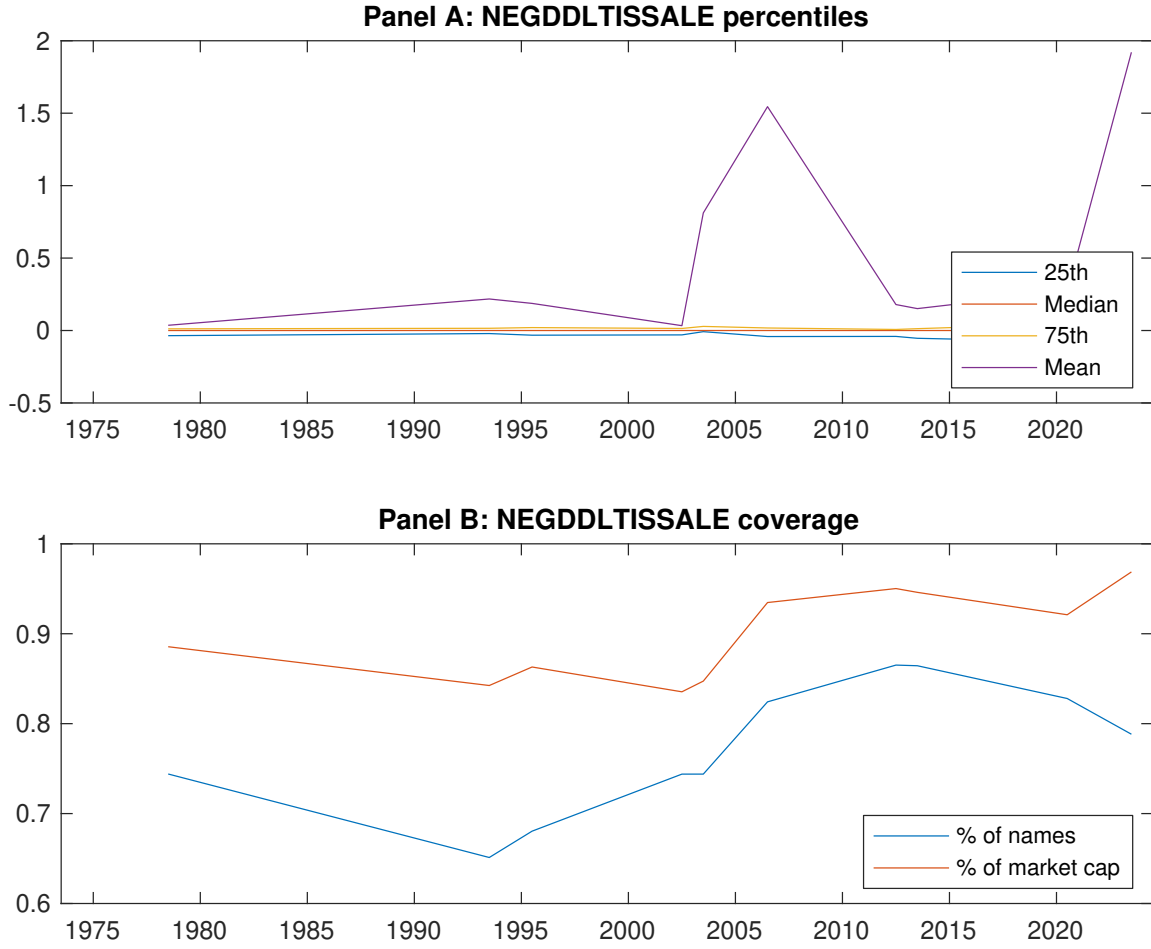


Figure 1: Times series of RDD percentiles and coverage. This figure plots descriptive statistics for RDD. Panel A shows cross-sectional percentiles of RDD over the sample. Panel B plots the monthly coverage of RDD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on RDD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on RDD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.62]	0.66 [3.70]	0.65 [3.25]	0.75 [4.24]	0.77 [3.97]	0.23 [3.27]
α_{CAPM}	-0.16 [-3.09]	0.06 [1.14]	-0.02 [-0.38]	0.15 [3.20]	0.11 [2.16]	0.27 [3.96]
α_{FF3}	-0.20 [-3.83]	0.02 [0.45]	0.05 [0.85]	0.14 [2.92]	0.09 [1.83]	0.29 [4.22]
α_{FF4}	-0.17 [-3.21]	0.05 [0.98]	0.09 [1.57]	0.09 [1.97]	0.08 [1.57]	0.25 [3.58]
α_{FF5}	-0.14 [-2.69]	-0.05 [-1.07]	0.08 [1.46]	0.04 [0.84]	0.06 [1.18]	0.20 [2.96]
α_{FF6}	-0.12 [-2.33]	-0.03 [-0.53]	0.11 [1.95]	0.01 [0.22]	0.06 [1.09]	0.18 [2.62]
Panel B: Fama and French (2018) 6-factor model loadings for RDD-sorted portfolios						
β_{MKT}	1.06 [87.24]	0.97 [89.57]	0.99 [75.37]	0.97 [93.04]	1.02 [84.54]	-0.04 [-2.47]
β_{SMB}	0.02 [0.85]	-0.11 [-6.61]	-0.01 [-0.27]	0.00 [0.27]	0.04 [2.44]	0.03 [1.21]
β_{HML}	0.16 [6.75]	0.10 [4.65]	-0.16 [-6.28]	-0.03 [-1.26]	-0.04 [-1.61]	-0.20 [-6.41]
β_{RMW}	-0.04 [-1.61]	0.16 [7.42]	0.01 [0.26]	0.14 [6.67]	-0.03 [-1.08]	0.01 [0.41]
β_{CMA}	-0.17 [-4.79]	0.07 [2.28]	-0.13 [-3.34]	0.18 [5.81]	0.18 [5.05]	0.35 [7.53]
β_{UMD}	-0.03 [-2.34]	-0.04 [-3.69]	-0.04 [-3.40]	0.04 [4.28]	0.01 [0.55]	0.03 [2.22]
Panel C: Average number of firms (n) and market capitalization (me)						
n	662	526	1031	582	637	
me (\$10 ⁶)	1718	2811	2215	2901	1687	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the RDD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.23 [3.27]	0.27 [3.96]	0.29 [4.22]	0.25 [3.58]	0.20 [2.96]	0.18 [2.62]
Quintile	NYSE	EW	0.16 [3.39]	0.18 [3.85]	0.18 [3.78]	0.17 [3.69]	0.16 [3.37]	0.16 [3.41]
Quintile	Name	VW	0.24 [3.48]	0.29 [4.16]	0.31 [4.45]	0.25 [3.66]	0.24 [3.50]	0.21 [3.01]
Quintile	Cap	VW	0.23 [3.37]	0.27 [4.03]	0.29 [4.35]	0.24 [3.57]	0.19 [2.84]	0.16 [2.39]
Decile	NYSE	VW	0.31 [3.35]	0.38 [4.20]	0.38 [4.15]	0.31 [3.43]	0.25 [2.82]	0.22 [2.42]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.46]	0.21 [3.06]	0.23 [3.27]	0.20 [2.93]	0.16 [2.27]	0.13 [2.01]
Quintile	NYSE	EW	-0.10 [-1.67]					
Quintile	Name	VW	0.19 [2.68]	0.22 [3.26]	0.24 [3.48]	0.21 [3.08]	0.19 [2.73]	0.16 [2.41]
Quintile	Cap	VW	0.18 [2.61]	0.22 [3.31]	0.24 [3.56]	0.21 [3.16]	0.16 [2.31]	0.13 [2.00]
Decile	NYSE	VW	0.24 [2.58]	0.31 [3.31]	0.30 [3.26]	0.27 [2.89]	0.20 [2.18]	0.17 [1.90]

Table 3: Conditional sort on size and RDD

This table presents results for conditional double sorts on size and RDD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on RDD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high RDD and short stocks with low RDD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	RDD Quintiles					RDD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.63 [2.25]	0.98 [3.62]	0.97 [3.63]	1.09 [3.81]	0.70 [2.54]	0.07 [0.91]	0.09 [1.10]	0.08 [0.96]	0.06 [0.80]	0.08 [1.02]	0.08 [0.92]
	(2)	0.74 [2.79]	0.94 [3.73]	0.84 [3.35]	0.93 [3.77]	0.80 [3.19]	0.06 [0.79]	0.10 [1.28]	0.07 [0.86]	0.07 [0.90]	0.04 [0.55]	0.05 [0.62]
	(3)	0.74 [3.00]	0.83 [3.75]	0.85 [3.53]	0.89 [3.90]	0.84 [3.61]	0.10 [1.31]	0.15 [1.91]	0.16 [2.08]	0.13 [1.62]	0.14 [1.83]	0.12 [1.51]
	(4)	0.70 [3.13]	0.82 [3.86]	0.83 [3.70]	0.80 [3.82]	0.84 [3.93]	0.15 [1.92]	0.17 [2.25]	0.17 [2.20]	0.14 [1.82]	0.13 [1.62]	0.11 [1.39]
	(5)	0.42 [2.12]	0.65 [3.62]	0.60 [3.00]	0.70 [3.87]	0.71 [3.78]	0.29 [3.18]	0.33 [3.57]	0.34 [3.69]	0.28 [3.08]	0.21 [2.32]	0.18 [1.99]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	RDD Quintiles					RDD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	382	384	383	383	379	36	33	31	32	34	
	(2)	106	106	106	106	106	60	60	59	60	59	
	(3)	76	76	76	76	76	107	108	104	106	106	
	(4)	64	64	65	65	64	230	238	229	235	228	
(5)	59	59	59	59	59	1490	2124	1805	2213	1544		

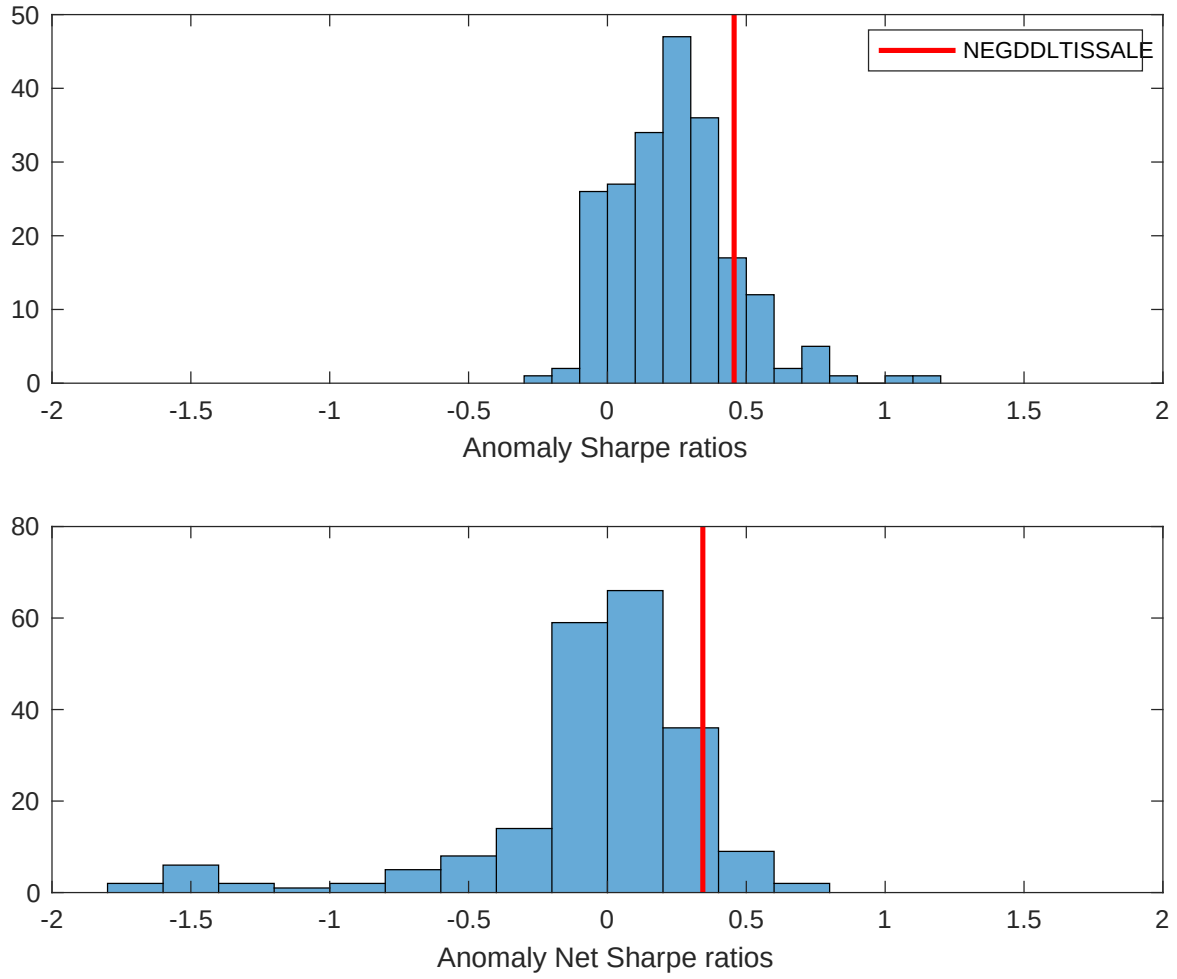


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the RDD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

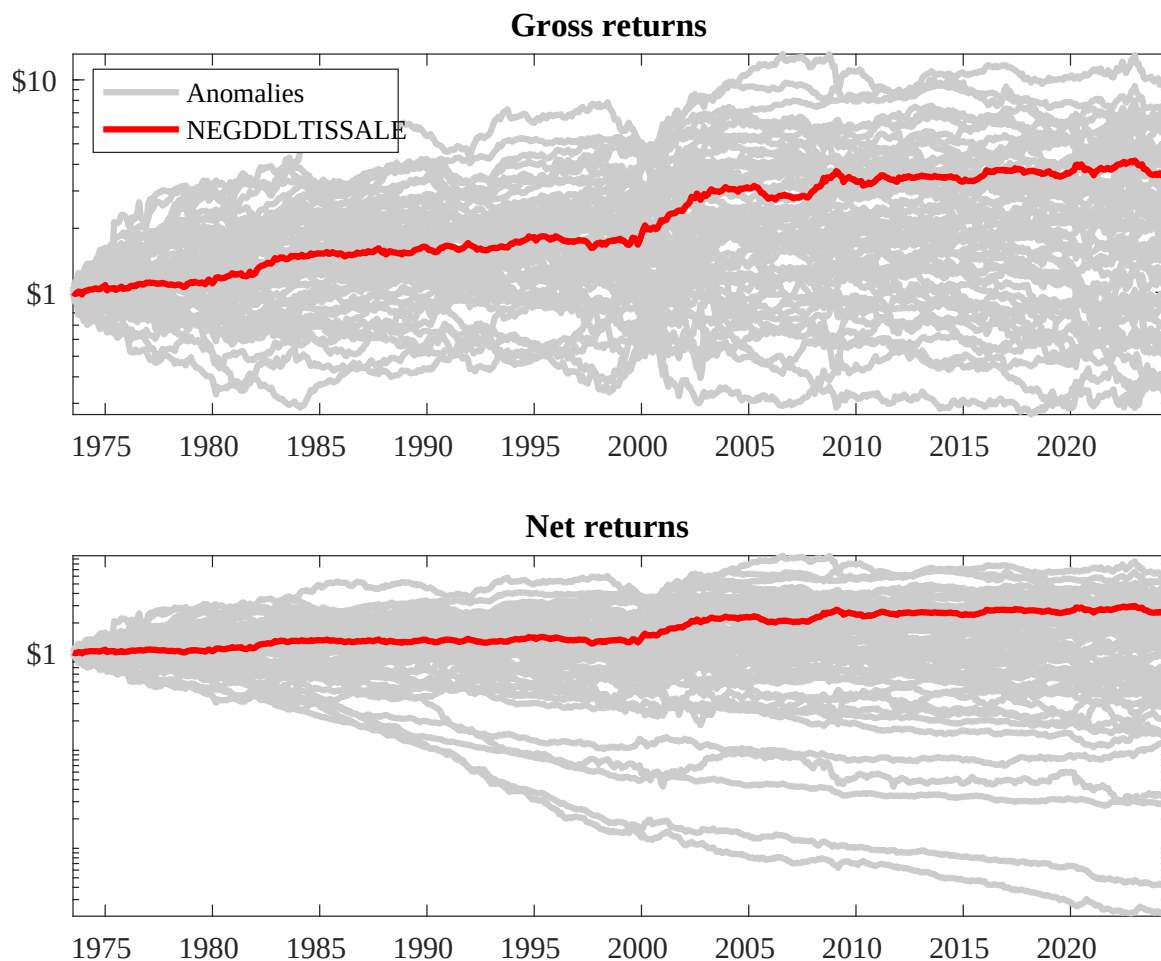
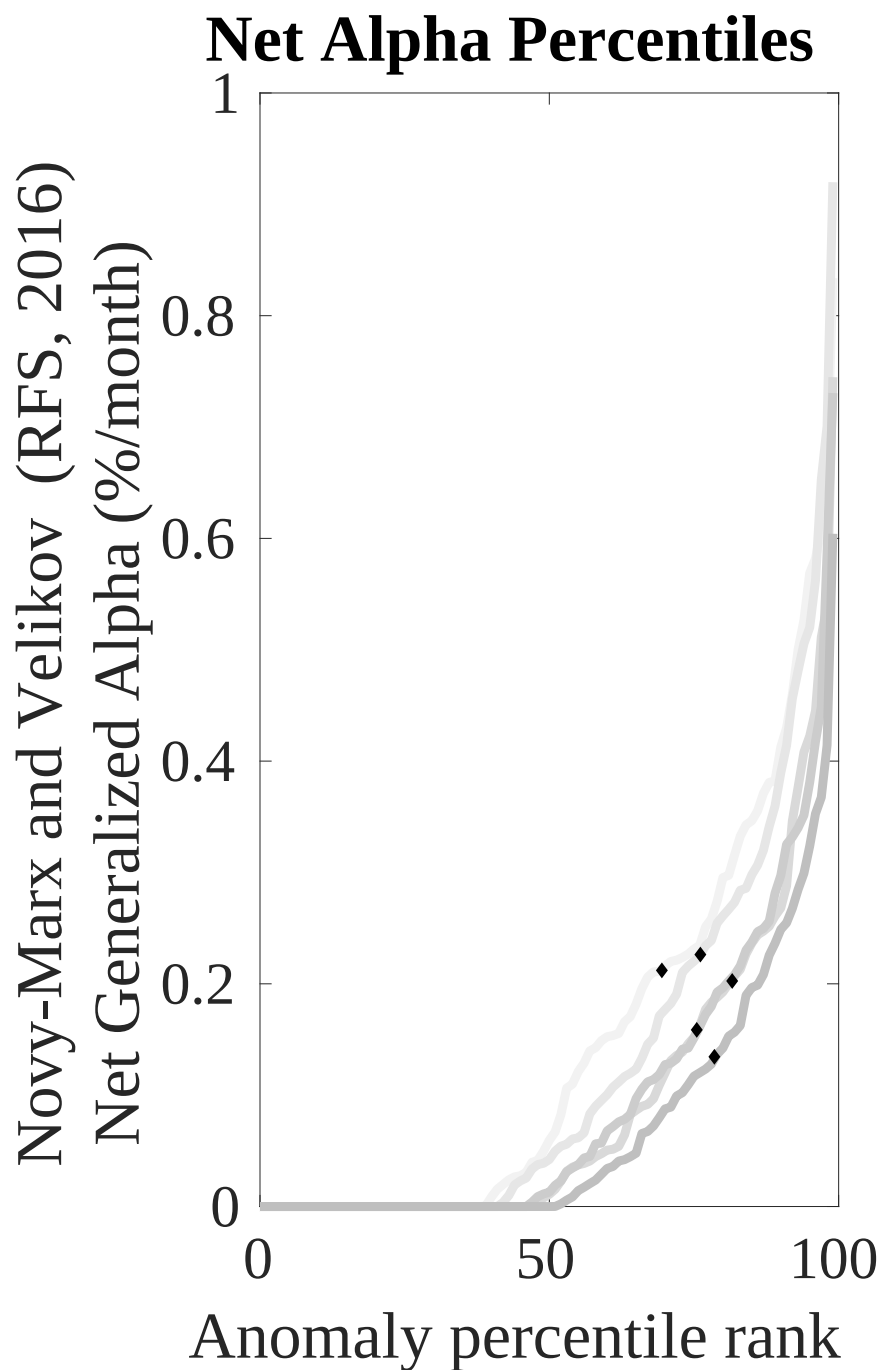
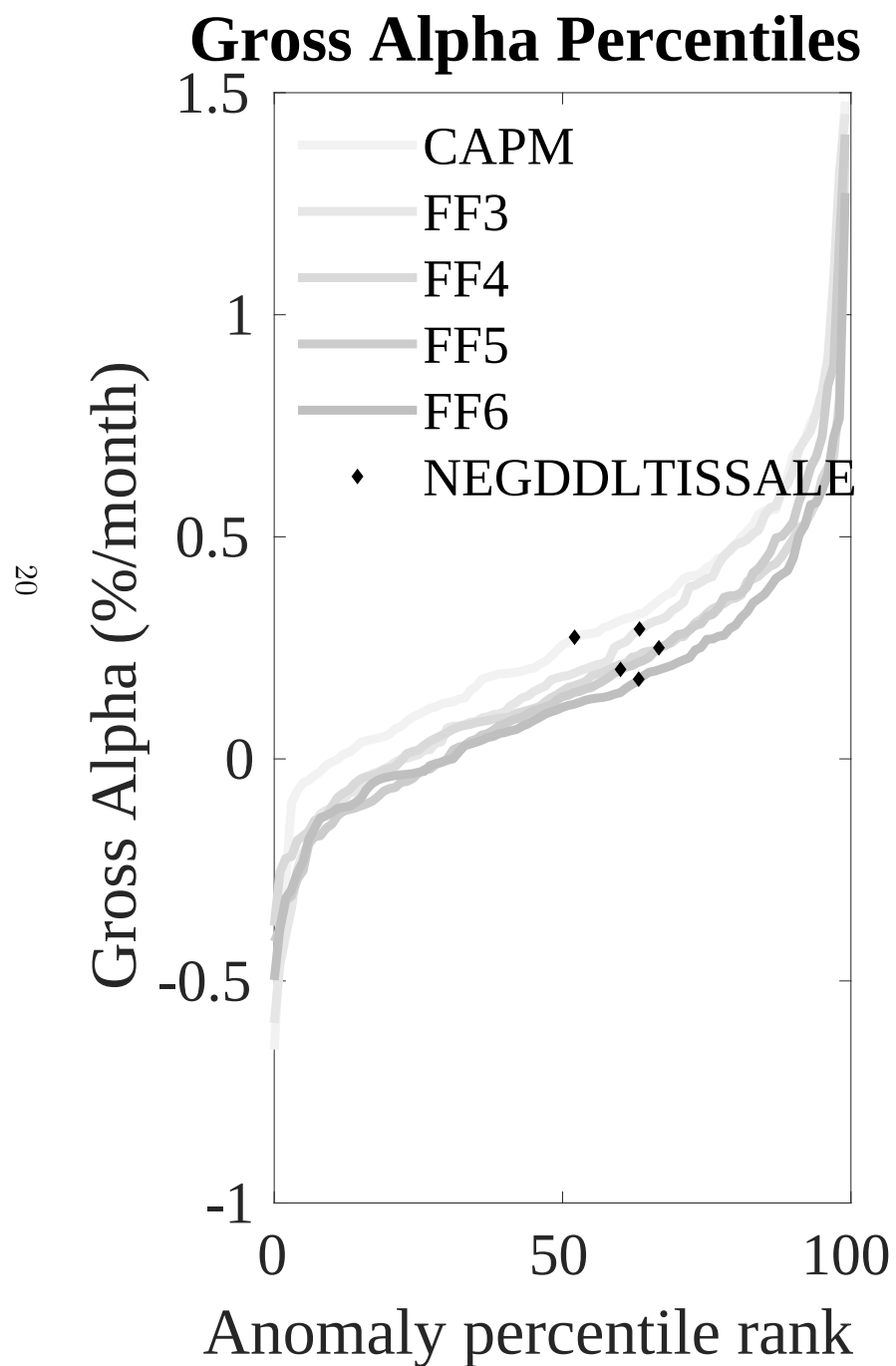


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the RDD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



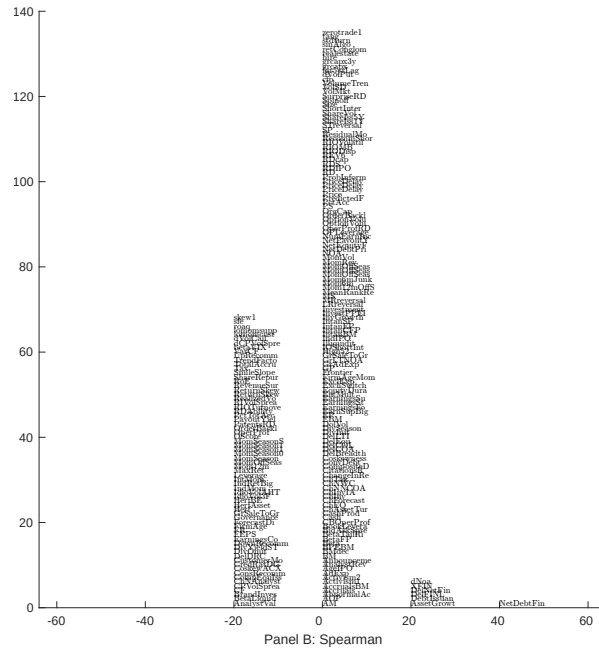
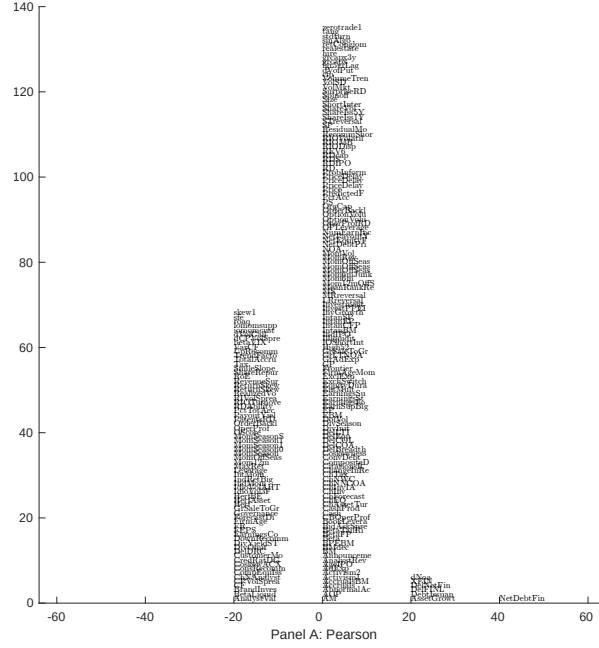


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with RDD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

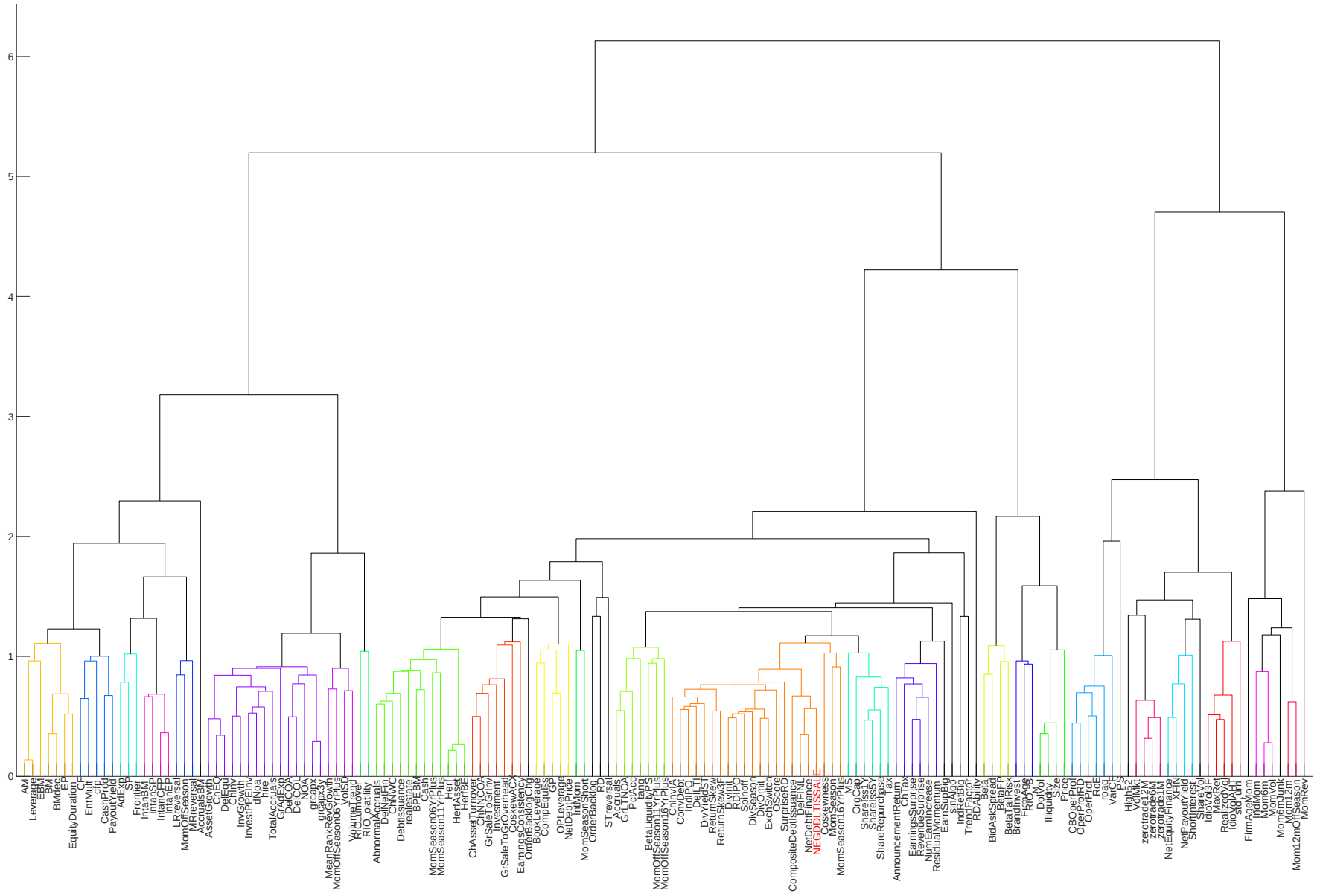


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

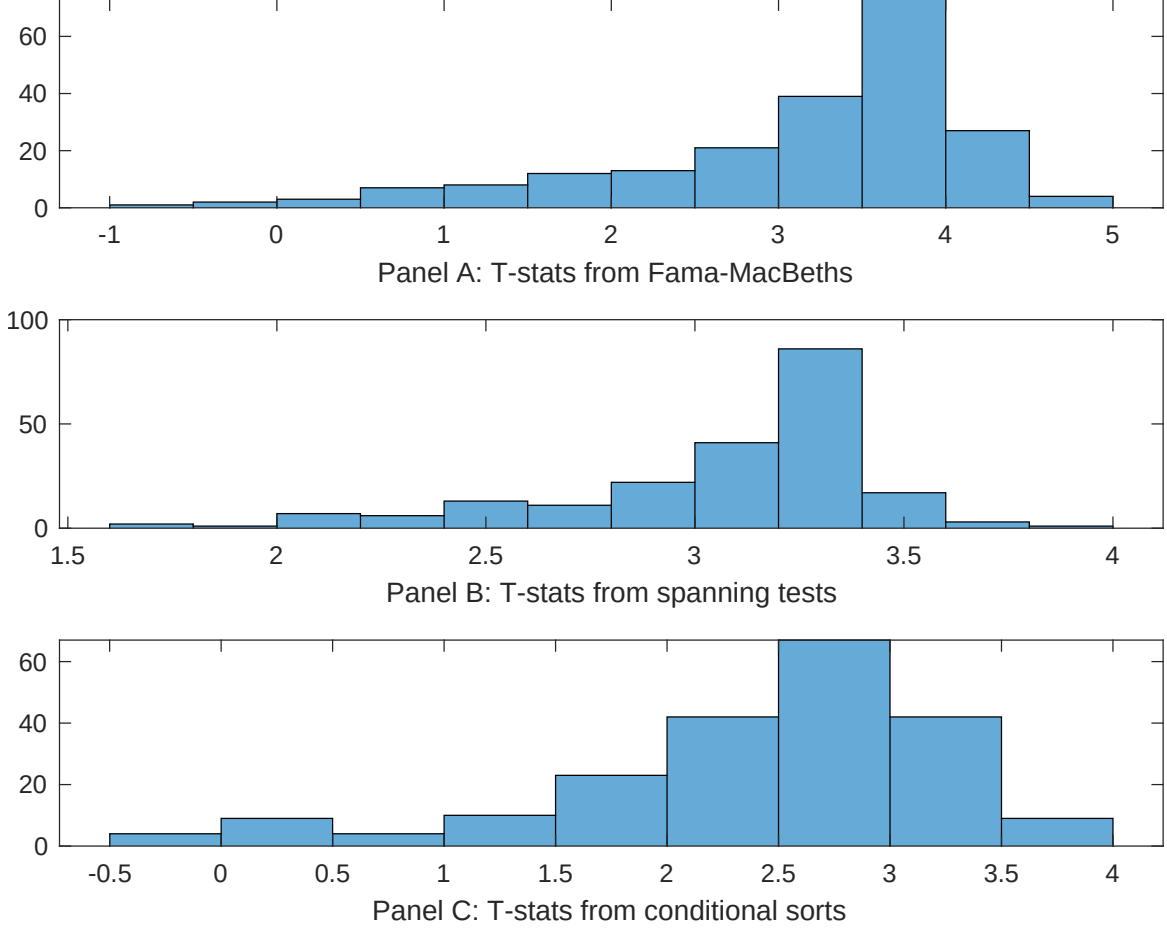


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of RDD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{RDD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{RDD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on RDD. Stocks are finally grouped into five RDD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted RDD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on RDD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{RDD}RDD_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.42]	0.13 [5.46]	0.14 [5.70]	0.14 [5.32]	0.14 [5.89]	0.13 [5.42]	0.15 [5.68]
RDD	0.95 [0.14]	0.89 [0.01]	0.68 [1.08]	0.29 [3.55]	-0.13 [-0.19]	0.19 [2.95]	-0.54 [-0.66]
Anomaly 1	0.21 [9.63]						0.96 [1.45]
Anomaly 2		0.18 [9.44]					-0.53 [-1.14]
Anomaly 3			0.20 [6.75]				0.98 [1.68]
Anomaly 4				0.40 [6.87]			0.91 [1.63]
Anomaly 5					0.11 [8.82]		0.76 [3.66]
Anomaly 6						0.94 [6.22]	0.81 [0.58]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the RDD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{RDD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Inventory Growth, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15 [2.28]	0.15 [2.22]	0.15 [2.23]	0.17 [2.54]	0.17 [2.45]	0.19 [2.73]	0.15 [2.24]
Anomaly 1	20.41 [5.60]						11.97 [2.40]
Anomaly 2		17.17 [4.51]					7.28 [1.45]
Anomaly 3			14.69 [4.36]				11.23 [3.12]
Anomaly 4				6.28 [2.37]			4.45 [1.63]
Anomaly 5					5.44 [1.38]		-1.81 [-0.44]
Anomaly 6						6.94 [1.89]	3.61 [0.94]
mkt	-3.56 [-2.33]	-3.21 [-2.07]	-1.59 [-0.98]	-3.72 [-2.38]	-3.61 [-2.30]	-3.42 [-2.18]	-1.83 [-1.13]
smb	1.21 [0.52]	1.25 [0.53]	7.73 [3.00]	3.70 [1.55]	2.62 [1.10]	3.48 [1.46]	5.67 [2.08]
hml	-18.60 [-6.29]	-18.29 [-6.12]	-17.80 [-5.92]	-19.62 [-6.50]	-19.67 [-6.49]	-20.76 [-6.69]	-18.02 [-5.90]
rmw	-0.80 [-0.26]	-0.10 [-0.03]	-7.33 [-2.01]	2.36 [0.76]	1.47 [0.48]	1.76 [0.57]	-6.43 [-1.76]
cma	30.13 [6.57]	29.67 [6.31]	25.85 [5.13]	29.83 [5.80]	28.82 [4.35]	29.23 [5.19]	17.62 [2.56]
umd	2.36 [1.53]	2.19 [1.40]	3.58 [2.33]	3.04 [1.92]	3.80 [2.43]	3.23 [2.05]	1.45 [0.92]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	17	16	15	14	14	20

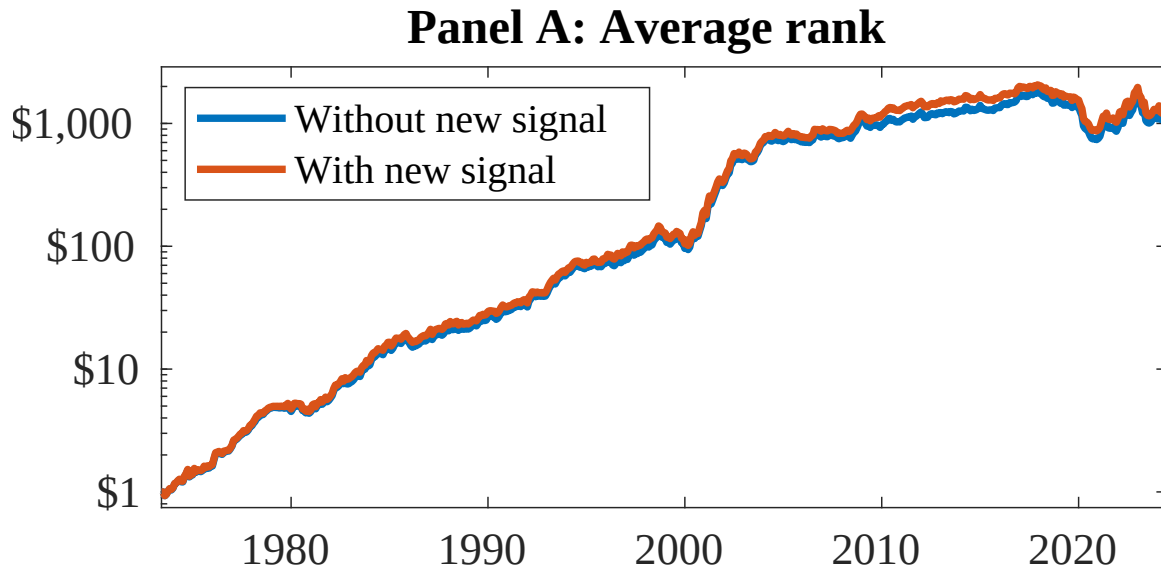


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as RDD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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